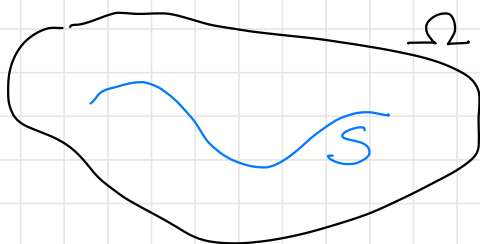


Teorema d'unicità del prolungamento analitico

Sia Ω aperto $\subseteq \mathbb{C}$ CONNESSO, e sia $S \subseteq \Omega$ tale che $\text{acc}(S) \cap \Omega \neq \emptyset$



Data $f: S \rightarrow \mathbb{C}$, esiste al più una funzione $\tilde{f}: \Omega \rightarrow \mathbb{C}$ olomorfa (analitica)

$$\text{tale che } \underbrace{\tilde{f}(x) = f(x) \quad \forall x \in S}_{\tilde{f}|_S = f}$$

Dim. Siano $\tilde{f}_1$ e $\tilde{f}_2$ due prolungamenti olomorfi di f . Devo mostrare $\tilde{f}_1 \equiv \tilde{f}_2$.

Considero $g := \tilde{f}_1 - \tilde{f}_2$. $\tilde{g}$ olomorfa.

g si annulla su S ($g|_S = \tilde{f}_1|_S - \tilde{f}_2|_S = f - f = 0$)

$$S \subseteq \mathcal{Z}(g)$$

Siccome per ip. $\Omega \cap \text{acc}(S) \neq \emptyset$

$\implies g \equiv 0$ su $\Omega \implies \tilde{f}_1 \equiv \tilde{f}_2$ su Ω . $\square$

PRINCIPIO ID.

Esempi di applicazioni

$$1) \quad \sin^2 z + \cos^2 z = 1 \quad \forall z \in \mathbb{C}$$

$$2) \quad \sin(2z) = 2 \sin z \cos z \quad \forall z \in \mathbb{C}$$

.....

Dim. 1)

$$\sin^2 x + \cos^2 x = 1 \quad \forall x \in \mathbb{R}$$

$$f(x) = \sin^2 x + \cos^2 x - 1 \equiv 0 \quad \forall x \in \mathbb{R}$$



$\Rightarrow$ $\exists$ almeno un prolungamento olomorfo

$$\nearrow \quad \tilde{f}_1(z) = \sin^2 z + \cos^2 z - 1 \quad \forall z \in \mathbb{C}$$

$$\searrow \quad \tilde{f}_2(z) = 0 \quad \forall z \in \mathbb{C}$$

$$\Rightarrow \quad \tilde{f}_1(z) = \tilde{f}_2(z) \quad \forall z \in \mathbb{C}$$

□

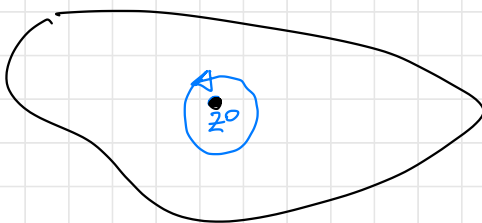
Notazione : calcolo di integrali in campo complesso (su circuiti)
(anche in campo reale)

- f olomorfa su $\Omega \subseteq \mathbb{C} \Rightarrow \oint f(z) dz = 0$
 $\nwarrow$ circuito a un punto

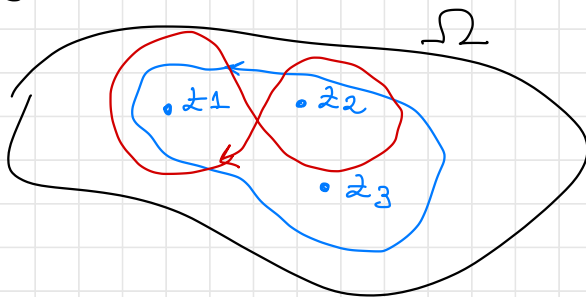
- f singolarità isolata in $z^0 \Rightarrow$

$$\oint_{\Gamma(z^0)} f(z) dz = 2\pi i \operatorname{Res}(f, z^0)$$

$\underbrace{\quad}_{\substack{\uparrow \\ \Omega}}$



- Più ingenua ???



Def. e calcolo di residui

Sia z^0 sing. isolato per f

$\text{Res}(f, z^0) := c_{-1}$ nello sviluppo di Laurent di f di centro z^0 .

- z^0 sing. eliminabile $\Rightarrow \text{Res}(f, z^0) = 0$
- z^0 sing. essenziale $\Rightarrow$ non c'è modo diretto (bisogna cercarsi sviluppo Laurent).
- z^0 polo di ordine $\nu \Rightarrow$

$$\text{Res}(f, z^0) = \lim_{z \rightarrow z^0} \frac{1}{(\nu-1)!} D^{\nu-1} \left[(z-z^0)^\nu f(z) \right]$$

In particolare se $\nu=1$ (polo semplice)

$$\text{Res}(f, z^0) = \lim_{z \rightarrow z^0} [(z-z^0) f(z)]$$

Dim. polo semplice

$$f(z) = \frac{c_{-1}}{(z-z^0)} + c_0 + c_1(z-z^0) + o(z-z^0)$$

$$(z-z^0) f(z) = c_{-1} + \underbrace{c_0(z-z^0) + c_1(z-z^0)^2 + o(z-z^0)^2}_0$$

$$\lim_{z \rightarrow z^0} [(z-z^0) f(z)] = c_{-1}.$$

□

Esempi

1) $f(z) = \frac{\sin z}{z}$ $z^0 = 0$ eliminabile

$$\operatorname{Res}\left(\frac{\sin z}{z}, 0\right) = 0$$

2) $f(z) = e^{\frac{1}{z}}$ $z^0 = 0$ essenziale

$$\forall z \in \mathbb{C} \setminus \{0\} \quad e^{\frac{1}{z}} = \sum_{n=0}^{\infty} \frac{1}{n!} \cdot \frac{1}{z^n}$$

$$\operatorname{Res}(e^{\frac{1}{z}}, 0) = 1$$

$$w = \frac{1}{z}$$

$f(z) = e^{\frac{1}{z^2}}$ $z^0 = 0$ essenziale

$$\forall z \in \mathbb{C} \setminus \{0\} \quad e^{\frac{1}{z^2}} = \sum_{n=0}^{\infty} \frac{1}{n!} \cdot \frac{1}{z^{2n}}$$

$$(e^w = \sum_{n=0}^{\infty} \frac{1}{n!} w^n)$$

$$w = \frac{1}{z^2}$$

$$\operatorname{Res}(e^{\frac{1}{z^2}}, 0) = 0$$

3) $f(z) = \frac{1}{z^{\nu}}$ $z^0 = 0$ polo di ordine ν

$$\nearrow \nu = 1$$

$$\operatorname{Res}\left(\frac{1}{z}, 0\right) = 1$$

$$\searrow \nu \neq 1$$

$$\operatorname{Res}\left(\frac{1}{z^{\nu}}, 0\right) = 0$$

$$4) f(z) = \frac{\sin z}{z^3} \quad z_0 = 0 \quad \text{polo ordine 2}$$

$$\lim_{z \rightarrow 0} 2 \cdot \frac{\sin z}{z^3} = \lim_{z \rightarrow 0} \frac{\sin z}{z} \in \mathbb{C} \setminus \{0\}$$

$$\Rightarrow \text{Res}(f, 0) = \lim_{z \rightarrow 0} \mathcal{D} \left[\underbrace{z^2 f(z)}_{\frac{\sin z}{z}} \right]$$

$$\lim_{z \rightarrow z_0} \frac{1}{(r-1)!} \mathcal{D}^{r-1} \left[(z-z_0)^r f(z) \right]$$

$$= \lim_{z \rightarrow 0} \frac{\cos z \cdot z - \sin z}{z^2} \stackrel{[H]}{=} \lim_{z \rightarrow 0} \frac{\cancel{\cos z} - z \sin z - \cancel{\cos z}}{2z}$$

$$= -\frac{1}{2} \lim_{z \rightarrow 0} \sin z = 0$$

$$5) f(z) = z \cot z = \frac{z}{\sin z} \cos z$$

$$\sin z = 0 \iff z_k = k\pi \quad k \in \mathbb{Z}$$

Caso $k=0$ $z_0=0$ sing. eliminabile

$$f(z) = \left(\frac{z}{\sin z} \right) \cdot \left(\cos z \right)$$

$\searrow \quad \searrow$
 $1 \quad 1$

$$\Rightarrow \text{Res}(f, 0) = 0$$

Caso $k \neq 0$ z_k polo semplice

$$\lim_{z \rightarrow k\pi} (z - k\pi) \cdot f(z) = \lim_{z \rightarrow k\pi} (z - k\pi) \frac{\pm \cos z}{\sin z}$$

$k\pi$ $(-1)^k$

$$= \lim_{w \rightarrow 0} \frac{w}{\sin(w + k\pi)} \cdot (k\pi) \cancel{(-1)^k}$$

$\uparrow$
 $w = z - k\pi$
 $w \rightarrow 0$

$$\sin w \cdot \underbrace{\cos(k\pi)}_{(-1)^k} + \cancel{\cos w \sin(k\pi)}$$

$$= \lim_{w \rightarrow 0} \left(\frac{w}{\sin w} \right) (k\pi) = k\pi$$

$\text{Res}(f, z_k) = k\pi \quad \forall k \in \mathbb{Z}$

6) Osservazione:

$$f = \frac{g}{h} \quad g \text{ olomorfo, } h \text{ zero di ord 1 in } z^0 \\ (h(z^0)=0, h'(z^0) \neq 0)$$

$$\Rightarrow \boxed{\operatorname{Res} \left(\frac{g}{h}, z^0 \right) = \frac{g(z^0)}{h'(z^0)}} \quad (*)$$

$$5) f(z) = \frac{z \cos z}{\sin z} \Rightarrow \operatorname{Res}(f, z_k) = \frac{z_k \cos z_k}{-\sin z_k} \quad \begin{matrix} K\pi \\ \parallel \end{matrix}$$

dim (*):

$$\begin{cases} \rightarrow g(z^0) \neq 0 \text{ caso 1) } \\ \rightarrow g(z^0) = 0 \text{ caso 2) } \end{cases}$$

caso 1 z^0 polo semplice

$$\frac{(z-z^0) g(z)}{h(z)} = \frac{g(z^0) \overset{\neq 0}{(z-z^0)} + o(z-z^0)}{\underbrace{h'(z^0)}_{\neq 0} (z-z^0) + o(z-z^0)} \xrightarrow{z \rightarrow z^0} \frac{g(z^0)}{h'(z^0)} \quad \begin{matrix} \in \mathbb{C} \setminus \{0\} \\ \cup \end{matrix}$$

caso 2 z^0 zing. eliminabile

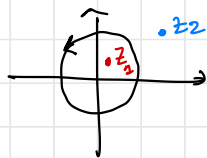
$$\frac{g(z)}{h(z)} = \frac{g'(z^0)(z-z^0) + o(z-z^0)}{h'(z^0)(z-z^0) + o(z-z^0)} \rightarrow \frac{g'(z^0)}{h'(z^0)} \in \mathbb{C}$$

Def e esodo indice di avvolgimento

Def "intuitiva" Sia γ circuito $\subseteq \mathbb{C}$ e sia $z^0 \notin \gamma$
Si definisce indice di avvolgimento di γ rispetto a z^0
il numero di volte che γ "gira" attorno a z^0 ,
contate col segno $\begin{cases} + \\ - \end{cases}$ nel caso di verso $\begin{cases} \text{antiorario} \\ \text{orario} \end{cases}$.

Esempi

• $\gamma = \mathcal{C}_1(0)$ $r(t) = e^{it} \quad t \in [0, 2\pi]$



$$\text{Ind}(\gamma, z_1) = 1$$

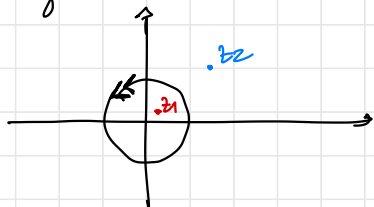
$$\text{Ind}(\gamma, z_2) = 0$$

• $\gamma = -\mathcal{C}_1(0)$



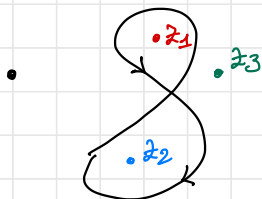
$$\begin{cases} \text{Ind}(\gamma, z_1) = -1 \\ \text{Ind}(\gamma, z_2) = 0 \end{cases}$$

• γ



$$r(t) = e^{it} \quad t \in [0, 4\pi]$$

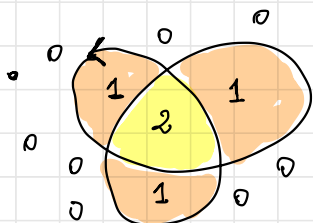
$$\begin{cases} \text{Ind}(\gamma, z_1) = 2 \\ \text{Ind}(\gamma, z_2) = 0 \end{cases}$$



$$\text{Ind}(\gamma, z_3) = 0$$

$$\text{Ind}(\gamma, z_1) = 1$$

$$\text{Ind}(\gamma, z_2) = -1$$



Def Se $\gamma \subseteq \mathbb{C}$ circolo, $z^0 \notin \gamma$, sia

$r(t): [a, b] \rightarrow \mathbb{C}$ parametrizzazione di γ ,

e sia $\rho(t) := |r(t) - z^0|$.

(Allora $\exists \theta: [a, b] \rightarrow \mathbb{C}$ tale che

$$\boxed{r(t) = z^0 + \rho(t) e^{i\theta(t)}} \quad)$$

$$\text{Ind}(\gamma, z^0) := \frac{\theta(b) - \theta(a)}{2\pi} \underset{(*)}{\in} \mathbb{Z}$$

$$(*) \quad r(a) = r(b)$$

$$\begin{aligned} r(a) &= z^0 + \underbrace{\rho(a)}_{=} e^{i\theta(a)} \\ r(b) &= z^0 + \underbrace{\rho(b)}_{=} e^{i\theta(b)} \end{aligned} \quad \left| \Rightarrow \begin{aligned} e^{i\theta(a)} &= e^{i\theta(b)} \\ \Downarrow \\ \theta(a) - \theta(b) &= \underbrace{k}_{\in \mathbb{Z}} \cdot 2\pi i \end{aligned} \right.$$

$$\rho(a) = |r(a) - z^0| = |r(b) - z^0| = \rho(b)$$

Formula per il calcolo dell'indice

$$\text{Ind}(\gamma, z^0) = \frac{1}{2\pi i} \int_{\gamma} \frac{1}{z - z^0} dz$$

(*)

- l'indice non cambia passando a una parametrizzazione equivalente.
- l'indice non cambia sostituendo γ con un circuito omotopo a γ (in $\mathbb{C} \setminus \{z^0\}$).

$$\star \quad r(t) = z^0 + \rho(t)e^{i\theta(t)} \quad t \in [a, b]$$

$$\int_{\gamma} \frac{1}{z - z^0} dz = \int_a^b \frac{\rho'(t)e^{i\theta(t)} + \rho(t) \cdot i\theta'(t)e^{i\theta(t)}}{\cancel{z^0} + \rho(t)e^{i\theta(t)} - \cancel{z^0}} dt$$

$$= \underbrace{\int_a^b \frac{\rho'(t)}{\rho(t)} dt}_{\log \rho(t) \Big|_a^b} + i \underbrace{\int_a^b \theta'(t) dt}_{[\theta(b) - \theta(a)]}$$

$$\frac{1}{2\pi i} \int_{\gamma} \frac{1}{z - z^0} dz = \frac{1}{2\pi i} \cdot i [\theta(b) - \theta(a)] = \text{Ind}(\gamma, z^0).$$

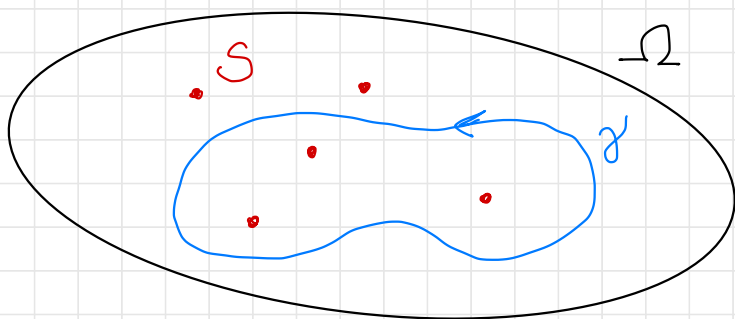
□

Teorema dei residui

Sia $\Omega \subseteq \mathbb{C}$ e sia $\gamma \subseteq \Omega$ un circuito omotopo a un punto in Ω

Sia $f: \Omega \setminus S \rightarrow \mathbb{C}$ olomorfa, dove S "insieme di singolarità" di f soddisfa

- $\text{acc}(S) \cap \Omega = \emptyset$
- $\gamma \subseteq \Omega \setminus S$



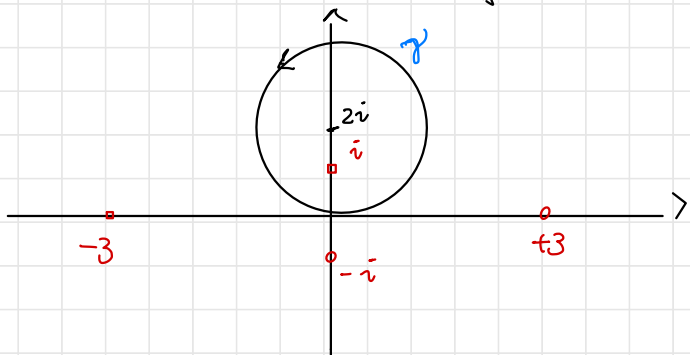
Allora:

$\text{Ind}(\gamma, z^0) \neq 0$ per al più un numero finito di punti, e vale:

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{z^0 \in S} \underbrace{\text{Res}(f, z^0)}_{\in \mathbb{C}} \cdot \underbrace{\text{Ind}(\gamma, z^0)}_{\in \mathbb{Z}}$$

Esempio standard

$$I = \int_{C_2(2i)} \frac{1}{\underbrace{(z^2+4)(z^2-9)}_f} dz = 2\pi i \cdot \text{Res}(f, i) \cdot \underbrace{\text{Ind}(\gamma, i)}_{+1}$$



$$\Omega = \emptyset$$

$$S = \{\pm i, \pm 3\}$$

$$\text{Res}(f, i) = \lim_{z \rightarrow i} (z - i) \cdot f(z) =$$

$$= \lim_{z \rightarrow i} \frac{1}{(z + i)(z^2 - 9)} =$$

$$= \frac{1}{2i} - \frac{1}{-10} = -\frac{1}{20i}$$

$$I = 2\pi i \left(-\frac{1}{20i} \right) = -\frac{\pi}{10}$$

Applicazioni del Teorema dei Residui in campo reale

Tipo 1

$$\int_0^{2\pi} f(\cos t, \sin t) dt = \int_0^{2\pi} \underbrace{g\left(\underbrace{e^{it}}_{r(t)}\right)}_{r'(t)} \underbrace{ie^{it}}_{r'(t)} dt = \int_{C_1(0)} g(z) dz =$$

$$= 2\pi i \sum_{\substack{z_0 \in S(g) \\ |z_0| < 1}} \text{Res}(g, z_0)$$

↑
seg. soddisfi ip. teo. residui
su $\Omega \supseteq C_1(0)$

$$\begin{cases} \cos t = \frac{e^{it} + e^{-it}}{2} \\ \sin t = \frac{e^{it} - e^{-it}}{2i} \end{cases}$$

Esempio

$$\int_0^{2\pi} \frac{1}{2 + \sin t} dt = \int_0^{2\pi} \frac{1}{2 + \frac{e^{it} - e^{-it}}{2i}} dt$$

$$= \int_0^{2\pi} \frac{1}{2 + \frac{e^{it} - e^{-it}}{2i}} \cdot \underbrace{ie^{it}}_{\leftarrow g(e^{it})} dt$$

$$= \int_0^{2\pi} \left[\frac{2i}{4i + e^{it} - e^{-it}} \cdot \frac{1}{e^{it}} \right] \cdot ie^{it} dt \quad \leftarrow g(e^{it})$$

$$\Rightarrow g(z) = \frac{2}{4i + z - z^{-1}} \cdot \frac{1}{z} = \frac{2}{z^2 + 4iz - 1}$$

g holomorphic on $\{z \neq \pm i\}$

$$z_{\pm} \text{ zero of } z^2 + 4iz - 1$$

TO BE CONTINUED ...